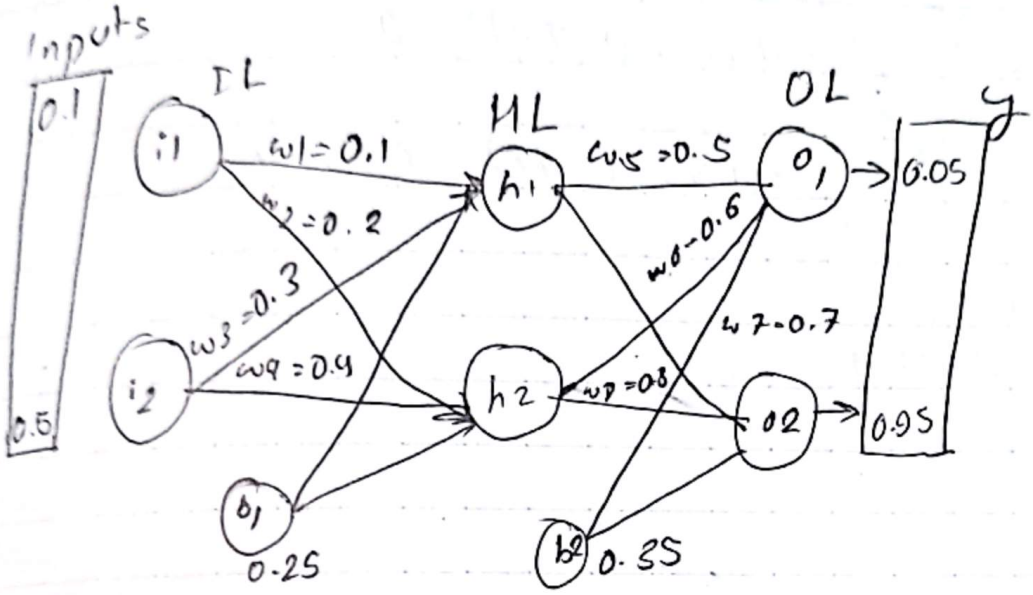
Solve Neural Network Below



Forward Pass

$z_1 = i_1 \cdot w_1 + i_2 \cdot w_3 + b_1$
$= 0.1 \times 0.1 + 0.5 \times 0.3 + 0.25$
$= 0.41$

$z_2 = i_2 \cdot w_2 + i_2 \cdot w_4 + 0.25$
$= 0.47$

$h_1 = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0.41}} = 0.60$

$h_2 = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0.47}} = 0.61$

$o1 = h_1 w_5 + h_2 w_6 + 0.35$
$= 0.60 \cdot 0.5 + 0.61 \cdot 0.6 + 0.35$
$= 1.016$

$\hat{y} = \frac{1}{1+e^{-a}} = \frac{1}{1+e^{-1.016}} = 0.73$

$o2 = h_1 w_7 + h_2 w_8 + b_2$
$= 0.60 \cdot 0.7 + 0.61 \cdot 0.8 + 0.35$
$= 1.258$

$\hat{y} = \frac{1}{1+e^{-02}} = \frac{1}{1+e^{-1.258}} = 0.78$

$E_{total} = \sum \frac{1}{2}(y - \hat{y})^2$

$$E_{o1} = \frac{1}{2}(y - \hat{y}_1)^2 = \frac{1}{2}(0.05 - 0.73)^2 = 0.23$$

$$E_{o2} = \frac{1}{2}(y_2 - \hat{y}_2)^2 = \frac{1}{2}(0.95 - 0.73) = 0.014$$

$$E_{total} = E_{o1} + E_{o2}$$
$$= 0.231 + 0.014$$
$$= 0.224$$

Backward Pass

$$w_5 = ?, \frac{\partial E_{total}}{\partial w_5} = \frac{\partial E_{total}}{\partial \hat{y}_1} \times \frac{\partial \hat{y}_1}{\partial o_1} \times \frac{\partial o_2}{w_5}$$

$$\frac{\partial E_{total}}{\partial \hat{y}_1} = \frac{\partial}{\partial \hat{y}_1} \frac{1}{2}(y_1 - \hat{y}_1)^2 + 0$$
$$= \frac{1}{2} \frac{\partial}{\partial \hat{y}_1} (y - \hat{y}_1)^2$$
$$= \frac{1}{2} \times 2 (y_1 - \hat{y}_1)^{2-1}$$
$$= (y_1 - \hat{y}_1)$$
$$= 0.05 - 0.73$$
$$= -0.68$$

$$\frac{\partial \hat{y}_1}{\partial o_1} = \frac{\partial}{\partial o_1} \left(\frac{1}{1+e^{-o}}\right)$$
$$= \frac{\partial}{\partial o_1} (1 + e^{-o_2})^{-1}$$
$$= \frac{e^{-o_1}}{(1+e^{-o_1})^2}$$
$$= \left(\frac{1}{1+e^{-o_1}}\right)^2 \times e^{-o_1}$$
$$= \hat{y}_1 (1 - \hat{y}_1)$$
$$= 0.73 (1 - 0.73)$$
$$= 7.83$$

$$\frac{\partial o1}{\partial w_3} = \frac{\partial}{\partial w}(h_1 w_5 + h_1 w_6 + 0.35)$$

$$= h_1 \cdot \frac{\partial w_5}{\partial w_5} = h_1 = 0.60$$

$$\frac{\partial E_{total}}{\partial \hat{y}_1} = (y_1 - \hat{y}_1) \times \hat{y}_1 (1 - \hat{y}_1) \times h_1$$

$$= -0.08 \times 0.73(1 - 0.73) \times 0.60$$

$$= 0.081$$

$$w_5 new = w_{5old} - \eta \frac{\partial E_{total}}{\partial w_5}$$

$$= 0.5 - 0.1 \times 0.08$$

$$= 0.492$$

$$w_6 new, \frac{\partial E_{total}}{\partial w_6} = \frac{\partial E_{total}}{\partial \hat{y}_1} \times \frac{\partial \hat{y}_1}{\partial o_1} \times \frac{\partial o_1}{\partial w_6}$$

$$\frac{\partial \hat{y}_1}{\partial o_1} = \hat{y}_1 (1 - \hat{y}_1)$$

$$\frac{\partial o_1}{\partial w_6} = \frac{\partial}{\partial w}(h_1 w_5 + h_2 w_6 + 0.35)$$

$$= h_2$$

$$\frac{\partial E_{total}}{\partial w_6} = (y - \hat{y}) \times \hat{y}(1 - \hat{y}) \times h_2$$

$$= (0.05 - 0.73) \times 0.73(1 - 0.73) \times 0.61$$

$$= -0.081$$

$$w_6 new = w_6 - \eta \frac{\partial E_{total}}{\partial w_6}$$

$$= 0.6 - (0.1 \times -0.081)$$

$$= 0.59$$

$$w_8, \frac{\partial E_{total}}{\partial w_8} = \frac{\partial E_{total}}{\partial y_2'} \times \frac{\partial y_2'}{\partial o_2} \times \frac{\partial o_2}{\partial w_8}$$

$$= -0.171 \times 0.1721 \times 0.615$$

$$= -0.018$$

$$w_8 new = w_8 - \frac{\partial E_{total}}{\partial w_8}$$

$$= 0.8 - (0.1 \times -0.018)$$

$$= 0.801$$

$$w_7 new = \frac{\partial E_{total}}{\partial w_7} = \frac{\partial E_{total}}{\partial y_2'} \times \frac{\partial y_2'}{\partial o_2} \times \frac{\partial o_2}{\partial w_7}$$

$$= 0.779 - 0.95 \times (0.779(1-0.779)) \times 0.68$$

$$= 0.171 \times 0.1721 \times 0.68$$

$$= -0.0176$$

$$w_7 new = w_7 old - \frac{\partial E_{total}}{\partial w_7}$$

$$= 0.7 - 0.1 \times (-0.176)$$

$$= 0.7 + 0.00176$$

$$= 0.7017$$

$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial y_1'} \times \frac{\partial y_1'}{\partial z_4} \times \frac{\partial z_4}{\partial h_2} \times \frac{\partial h_2}{\partial z_2} \times \frac{\partial z_2}{\partial w_2}$$

$$= (y_1 - y_2')(y_2(1-y_2)) \times w_8 new \times h_2(1-h_2)$$

$$= (0.95 - 0.779)(0.779(1-0.779)) \times 0.701 \times 0.65 \times (-0.615) \times 0.5$$

$$= 0.001789$$

$$w_1, \frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial y_1} \times \frac{\partial y_1}{\partial z_{23}} \times \frac{\partial z_{23}}{\partial h_{1,out}} \times \frac{\partial h_{1,out}}{\partial z_1} \times \frac{\partial z_1}{\partial w_1}$$

$$= (y_1 - \bar{y}_1)(y_1(1-y_1)) \times w_5 \times h_{1,out}(1-h_{1,out}) \times i_1$$

$$= 0.884 \times 0.1952 \times 0.499 \times 0.501 \times 0.399 \times 0.1$$

$$= 0.00157$$

$$w_2 = \frac{\partial E_{total}}{\partial w_2} = \frac{\partial E_{total}}{\partial y_1} \times \frac{\partial y_1}{\partial z_{23}} \times \frac{\partial z_{23}}{\partial h_{1,out}} \times \frac{\partial h_{1,out}}{\partial h_2} \times \frac{\partial z_1}{\partial w_2} \times i_2$$

$$= 0.884 \times 0.1952 \times 0.499 \times 0.501 \times 0.399 \times 0.5$$

$$= 0.0787$$

$$w_3 \frac{\partial E_{total}}{\partial w_3} = \frac{\partial E_{total}}{\partial y_2} \times \frac{\partial y_2}{\partial z_{24}} \times \frac{\partial z_{24}}{\partial h_{2,out}} \times \frac{\partial h_{2,out}}{\partial z_2} \times \frac{\partial z_2}{\partial w_3}$$

$$= (y_2 - \bar{y}_2)(y_2(1-y_2)) \times w_5 \times h_2(1-h_2)$$

$$= 0.000558.$$